

# Damien Robert

Deep Learning for Remote Sensing & Environment



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I am a postdoctoral researcher in the [EcoVision](#) lab at the [University of Zurich](#), where I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on “*Efficient Learning on Large-Scale 3D Point Clouds*” at [IGN](#) and [ENGIE](#), under the supervision of [Loic Landrieu](#) and [Bruno Vallet](#).

📍 3D point clouds   🌿 vegetation mapping   🦋 species distribution   🌐 remote sensing   ⚡ efficient ML

## ■ POSITIONS

- |           |                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–now  | <b>Postdoctoral Researcher</b><br><a href="#">EcoVision lab, University of Zurich</a> · Zurich, Switzerland<br>Deep learning for remote sensing and environment<br>PI: Jan Dirk Wegner |
| 2020–2024 | <b>PhD Student</b><br><a href="#">LASTIG, IGN / ENGIE Lab CRIGEN</a> · Paris, France<br>Efficient learning on large-scale 3D point clouds<br>Advisors: Loic Landrieu, Bruno Vallet     |
| 2017–2020 | <b>R&amp;D Engineer</b><br><a href="#">SIRADEL, ENGIE / ENGIE</a> · Rennes, France<br>Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data                   |
| 2017      | <b>Co-Founder</b><br><a href="#">Inspirama</a> · Rennes, France<br>Website gathering book recommendations from inspiring people                                                        |
| 2015      | <b>R&amp;D Intern</b><br><a href="#">Dassault Systèmes</a> · Providence, RI, USA<br>Dimensionality reduction and dynamic system modeling                                               |
| 2014      | <b>R&amp;D Intern</b><br><a href="#">Dassault Systèmes</a> · Providence, RI, USA<br>UX design                                                                                          |
| 2014      | <b>R&amp;D Intern</b><br><a href="#">Dassault Systèmes</a> · Providence, RI, USA<br>UX design                                                                                          |

## ■ EDUCATION

- |           |                                                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020–2024 | <b>PhD, Computer Science</b><br><a href="#">Université Gustave Eiffel</a> · Paris, France<br>Efficient Learning on Large-Scale 3D Point Clouds              |
| 2022      | <b>Summer School</b><br><a href="#">ICVSS</a> · Sicily, Italy<br>Computer vision courses by world-renowned experts in academia and industry                 |
| 2017      | <b>AI Fall School</b><br><a href="#">CNRS</a> · Lyon, France<br>Multi-disciplinary course for AI students and researchers                                   |
| 2011–2015 | <b>MSc, Engineering</b><br><a href="#">École Centrale de Lyon</a> · Lyon, France<br>Mathematics, Computer Science, Mechanics, Signal Processing, Automation |

## ■ AWARDS

- 2026 **Outstanding Reviewer** · [CVPR](#)
- 2025 **PhD Award accessit** · [AFRIF](#)  
[French Association for Pattern Recognition and Interpretation](#)
- 2024 **Outstanding Reviewer** · [ECCV](#)
- 2022 **Best Paper Finalist** · [CVPR](#)  
Top 0.4% of submissions

## ■ PUBLICATIONS

487 citations · h-index 7 · [Google Scholar](#), Sep 2026

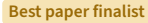
- 2026 **MALiBU3D: A Multitask Aerial LiDAR Benchmark for Large-Scale 3D Scene Understanding**  
L. Geist, E. Vincent, M. A. Chebbi, A. Garioud, N. Gonthier, S. Giordano, N. David, J. D. Wegner, L. Landrieu, **D. Robert**  
*Under review*
- FoM3D: Toward a foundation model for forest point clouds**  
Y. Yue, S. Puliti, **D. Robert**, A. Topaloglu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler  
*Under review*
- From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps**  
G. Sialelli, R. Young, Y. Jiang, ..., **D. Robert**, et al. (13 authors)  
*Under review*
- SAGE: A sampling-aware global evaluation benchmark for species distribution modeling**  
E. Arens, N. van Tiel, R. Zbinden, **D. Robert**, L. Drees, C. Vanalli, B. Kellenberger, N. Zimmermann, L. Pellissier, D. Tuia, J. D. Wegner  
*Under review*
- LitePT: Lighter Yet Stronger Point Transformer**  
Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler  
[CVPR](#) [Highlight](#)
- EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation**  
L. Geist, L. Landrieu, **D. Robert**  
[ICRA](#)
- FORMSPoT: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series**  
M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciaï  
[Remote Sensing of Environment](#)
- 2025 **MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation**  
K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner  
[PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science](#)
- SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology**  
E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann  
[CVPR workshop EarthVision](#)
- GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance**  
K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner  
[ISPRS Geospatial Week](#) [Oral](#)
- Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks**  
J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner  
[ICLR workshop Tackling Climate Change with Machine Learning](#)
- Lossy Neural Compression for Geospatial Analytics: A Review**  
C. Gomes, I. Wittmann, **D. Robert**, et al. (27 authors)  
[IEEE Geoscience and Remote Sensing Magazine](#)
- 2024 **Efficient Learning on Large-Scale 3D Point Clouds**  
**D. Robert**  
[PhD Thesis, Université Gustave Eiffel](#) [AFRIF PhD award accessit](#)  
Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loïc Landrieu, Bruno Vallet
- Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering**

D. Robert, H. Raguet, L. Landrieu  
3DV  · top 5.3% of submissions

2023 **Efficient 3D Semantic Segmentation with Superpoint Transformer**

D. Robert, H. Raguet, L. Landrieu  
ICCV · 26.8% acceptance rate

2022 **Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation**

D. Robert, B. Vallet, L. Landrieu  
CVPR   · top 0.4% of submissions

## ■ COLLABORATION & SUPERVISION

|                 |                                                                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Collaborators   | Konrad Schindler <a href="#">ETH</a> · Christian Rupprecht <a href="#">Oxford</a> · Jan Dirk Wegner <a href="#">UZH</a> · Loic Landrieu <a href="#">ENPC</a> · Vivien Sainte Fare Garnot <a href="#">UZH</a> · Lukas Drees <a href="#">UZH</a> · Martin Schwartz <a href="#">LSCE</a> · Elena Plekhanova <a href="#">WSL</a> |
| PhD Students    | Emilia Arens <a href="#">UZH</a> with Jan Dirk Wegner · Johannes Dollinger <a href="#">UZH</a> with Jan Dirk Wegner · Kaan Karaman <a href="#">UZH</a> with Jan Dirk Wegner · Louis Geist <a href="#">ENPC</a> with Loic Landrieu · Yuanwen Yue <a href="#">ETH</a> with Konrad Schindler and Christian Rupprecht            |
| Master Students | Valerio Schelbert <a href="#">WSL</a> · Jackson Sunny <a href="#">École Polytechnique</a> with SAMP                                                                                                                                                                                                                          |
| Advising        | UpCircle AI <a href="#">Switzerland</a> · École des Sciences Criminelles de Lausanne <a href="#">Switzerland</a> · SAMP AI <a href="#">France</a> · GEODIGITAL <a href="#">US</a>                                                                                                                                            |

## ■ REVIEWING, CHAIRING & ORGANISING

 Reviewer  Area chair  Organising committee

|      |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027 |  CVPR workshop EarthVision ·  3DV                                                                                                                                                                                                                     |
| 2026 |  ECCV ·  CVPR ·  EurIPS workshop REO                                                                                                                                 |
| 2025 |  CVPR ·  CVPR workshop EarthVision                                                                                                                                                                                                                  |
| 2024 |  ECCV ·  ISPRS Journal ·  ICLR workshop ML4RS ·  CVPR workshop EarthVision |
| 2023 |  CVPR ·  CVPR workshop EarthVision                                                                                                                                                                                                                |
| 2022 |  ISPRS Journal ·  CVPR workshop EarthVision                                                                                                                                                                                                       |

## ■ TEACHING

|      |                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------|
| 2026 | Transformer · <a href="#">UZH</a> · Lecture, Master 2                                                     |
| 2025 | AI4Good group projects · <a href="#">UZH</a> · Labs, Master 2                                             |
| 2025 | Transformer · <a href="#">UZH</a> · Lecture, Master 2                                                     |
| 2024 | Transformer, NeRFs, and Diffusion · <a href="#">UZH</a> · Lecture and labs, Master 2                      |
| 2023 | Deep Learning for Remote Sensing · <a href="#">ENSG-IGN</a> · Lecture, Master 2                           |
| 2022 | 3D Deep Learning for Remote Sensing · <a href="#">ISPRS'22</a> · Tutorial, Researchers                    |
| 2022 | 3D Deep Learning, Torch-Points3D & DeepViewAgg · <a href="#">ENGIE CRIGEN lab</a> · Tutorial, Researchers |
| 2022 | Deep Learning for Remote Sensing · <a href="#">ENSG-IGN</a> · Lecture, Master 2                           |
| 2020 | Deep Learning for Computer Vision · <a href="#">École Polytechnique</a> · Lecture, Master 1               |

## ■ OPEN-SOURCE SOFTWARE

|        |                                                                                         |                                                                                                                                                                    |
|--------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ 1055 |  136 | Superpoint Transformer ·  <a href="#">drprojects/superpoint_transformer</a>     |
| ★ 406  |  45  | LitePT ·  <a href="#">prs-eth/LitePT</a>                                        |
| ★ 237  |  24  | DeepViewAgg ·  <a href="#">drprojects/DeepViewAgg</a>                           |
| ★ 74   |  8   | Point Geometric Features ·  <a href="#">drprojects/point_geometric_features</a> |
| ★ 53   |  3   | NoRA ·  <a href="#">drprojects/nora</a>                                         |
| ★ 15   |  0   | torch-graph-components ·  <a href="#">drprojects/torch-graph-components</a>     |

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 |  <b>ESA-NASA workshop on AI Foundation Model for EO</b> · Building Earth Observation Embedding Workflows with TerraTorch · Virtual  |
|      |  <b>ESA Φ-lab</b> · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy                                |
| 2025 |  <b>MPI of Animal Behavior</b> · Research at EcoVision lab · Zurich, Switzerland                                                    |
|      |  <b>Global Forest Watch</b> · Geospatial foundation models for forest disturbance monitoring · Virtual                              |
|      |  <b>NatureFinance</b> · Research at EcoVision lab · Virtual                                                                         |
|      |  <b>NVIDIA Strategic Research Engagement</b> · Superpoint Transformer and EcoVision research · Zurich, Switzerland                  |
|      |  <b>CNES COMET TSI</b> · Remote Sensing for Species Distribution Modeling · Virtual                                                 |
| 2024 |  <b>3D Data Academy</b> · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube                               |
|      |  <b>Google DeepMind</b> · Research at EcoVision lab · Zurich, Switzerland                                                           |
|      |  <b>3DV'24</b> · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland                              |
| 2023 |  <b>École des Ponts, IMAGINE lab</b> · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France                            |
|      |  <b>National Land Survey of Finland (NLS)</b> · IGN's research on Large-Scale 2D and 3D Learning · Paris, France                    |
|      |  <b>ETH Zürich, Computer Vision and Geometry lab</b> · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland      |
|      |  <b>ETH Zürich, Photogrammetry and Remote Sensing lab</b> · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland |
|      |  <b>Valeo.ai</b> · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France                                   |
| 2022 |  <b>CV News</b> · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US                                |
|      |  <b>CVPR'22</b> · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US            |
|      | 45 talks given since 2021 — full list at <a href="https://drprojects.github.io/talks">drprojects.github.io/talks</a> .                                                                                               |

■ **SKILLS & LANGUAGES**

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                             |                                                                                                       |                                                                                                         |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Research  | <b>3D &amp; geometry</b> Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning<br><b>Efficient &amp; scalable learning</b> Efficient deep learning · Scalable model design · EO data representation and compression<br><b>Learning paradigms</b> Multimodal learning · Self-supervised learning · Representation learning<br><b>Environment &amp; Earth observation</b> Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology |                                                                                             |                                                                                                       |                                                                                                         |
| Tools     |  Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  PyTorch |  PyTorch Lightning |  PyTorch Geometric |
|           |  C++                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  Hydra   |  Weights & Biases  |  Git               |
|           |  Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SLURM                                                                                       |  scikit-learn      |  Plotly            |
|           |  Blender                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LaTeX                                                                                       |                                                                                                       |                                                                                                         |
| Languages | French Native                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | English Fluent                                                                              | Spanish Intermediate                                                                                  | German Beginner                                                                                         |
| Abroad    | Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                             |                                                                                                       |                                                                                                         |
| Interests | Outdoors · Climbing · Skateboarding · Basketball · Guitar · Drawing · Comics · Board games                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                             |                                                                                                       |                                                                                                         |